

ECSE 250

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March 20

Exam 2: After HW 6, April 3

Coverage: HWS 4-6

Topic: ① Expectation of functions of continuous RVs

② CDF/PDF of functions of continuous RVs

□ Expected values of functions of continuous RV

As before, we consider a function of RV X , given by $g(X)$, and its expectation is

$$E[g(X)] = \int_{-\infty}^{\infty} g(y) f_X(y) dy$$

Applications - Expectation/Variance of X^2

✓ Variance of RV X , e.g.,

$$\text{Var}(X) = E[(X - \underbrace{E[X]}_{\text{constant}})^2]$$

$$g(X) = (X - \mu)^2 \text{ Constant}$$

As is in discrete RV setting, we can have variance decomposition in continuous RV case,

$$\begin{aligned}\text{Var}(X) &= E[(X - E[X])^2] \\&= \int_{-\infty}^{+\infty} (x - E[X])^2 f_X(x) dx \\&= \int_{-\infty}^{+\infty} (x^2 + E[X]^2 - 2xE[X]) f_X(x) dx \\&= \int_{-\infty}^{+\infty} x^2 f_X(x) dx + E[X]^2 \int_{-\infty}^{+\infty} f_X(x) dx \\&\quad - 2E[X] \int_{-\infty}^{+\infty} x f_X(x) dx \\&= E[X^2] + E[X]^2 - 2E[X]^2 \\&= E[X^2] - E[X]^2 \leftarrow \text{Same as discrete RV.}\end{aligned}$$

① Start with linear function of X, Y

$$g(X, Y) = aX + bY$$

$\uparrow$ $\uparrow$
RV RV

$$i) E[g(x, y)] = a E[X] + b E[Y]$$

Using the Linearity of Expectation

$$g(x) = \underset{\substack{\uparrow \\ \text{RV}}}{X} + \underset{\substack{\uparrow \\ \text{constant}}}{c}$$

$$ii) E[g(x)] = E[X] + c$$

$$iii) g(x) = aX, \text{ given } \text{Var}(X)$$

$$\begin{aligned} \text{Var}(g(x)) &= E[g^2(x)] - E^2[g(x)] \\ &= E[a^2 X^2] - E^2[aX] \\ &= a^2 E[X^2] - a^2 E^2[X] \\ &= a^2 (E[X^2] - E^2[X]) \\ &= a^2 \underbrace{(E[X^2] - E^2[X])}_{\text{Var}(X)} \end{aligned}$$

Be careful,
this is not a $\text{Var}(X)$ →

$$= a^2 \text{Var}(X)$$

Example: Variance of a uniform continuous RV over $[a, b]$.
Answer: Use the variance decomposition

$$\rightarrow \text{Var}(X) = E[X^2] - E^2[X]$$

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{3(b-a)} x^3 \Big|_a^b$$

$$= \frac{1}{3} \cdot \frac{b^3 - a^3}{b-a}$$

$$E[X] = \int_{-\infty}^{+\infty} x \cdot \frac{1}{b-a} dx = \frac{a+b}{2}$$

$$\text{Plug in} \rightarrow \text{Var}(X) = \frac{b^3 - a^3}{3(b-a)} - \frac{(a+b)^2}{4}$$

$$= \frac{1}{12} (b-a)^2$$

Example: Variance of a Gaussian RV

Answer:

$$\text{PDF } f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

$$\text{Var}(X) = E[(X - \underbrace{E[X]}_{\mu})^2]$$

$$= \int_{-\infty}^{+\infty} (x - \mu)^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} dx$$

change of variable

$$x - \mu \leftrightarrow x \quad = \int_{-\infty}^{+\infty} x^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{x^2}{\sigma^2}} dx$$

Integration by parts

$$= -x\sigma^2 e^{-\frac{x^2}{2\sigma^2}} \Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} \sigma^2 e^{-\frac{x^2}{2\sigma^2}} dx$$

$$= 0 + \int_{-\infty}^{+\infty} \sigma^2 e^{-\frac{x^2}{2\sigma^2}} dx$$

$$= \sigma^2 \quad \leftarrow \text{Verify that the parameter appearing in the PDF is the standard deviation}$$

→ Homework: Variance of Exponential RV.

□ CDF/PDF of function of continuous RVs

We have a new continuous RV $Y = g(X)$.

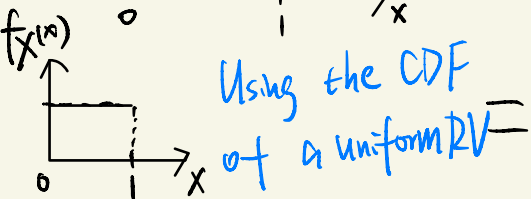
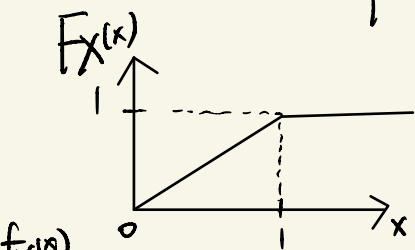
We know the PDF or CDF of RV X , and we want to find the PDF/CDF of the new RV Y , $F_Y(y)$ or $f_Y(y)$.

Example: X denotes a uniform RV $[0, 1]$.

Let $Y = g(X) = e^X$. What is CDF/PDF of Y ?

Answer: Based on definition,

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(e^X \leq y) \\ &= P(X \leq \ln y) \end{aligned}$$



Using the CDF

of a uniform RV =

$$= \begin{cases} \ln y, & \text{if } \ln y \in [0, 1] \\ 1, & \text{if } \ln y > 1 \\ 0, & \text{if } \ln y < 0 \end{cases}$$

$$= \begin{cases} \ln y, & \text{if } y \in [e^0, e'] \\ 1, & \text{if } y \geq e' \\ 0, & \text{if } y \leq e^0 \end{cases}$$

Based on the CDF of Y , we can obtain

$$\text{PDF } f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$= \begin{cases} \frac{1}{y}, & \text{if } y \in [e^0, e'] \\ 0, & \text{if } y > e' \text{ or } < e^0 \end{cases}$$

To generalize this steps, what we did is like:

Special case 1 If $Y = g(X)$ and g is invertible (1 to 1 mapping) and both X and Y are positive, then

$$\begin{aligned} \text{CDF of } Y \text{ will be } F_Y(y) &= P(Y \leq y) = P(X \leq g^{-1}(y)) \\ &= F_X(g^{-1}(y)) \end{aligned}$$

Using Invertible of $g(x)$
← Composition of $F_X \circ g^{-1}$

PDF of Y $f_Y(y) = \frac{d F_Y(y)}{d y}$

(Using Chain rule) $= f_X(g^{-1}(y)) \cdot \frac{d}{d y} g^{-1}(y)$

Special case 2

A linear function/transformation

Let RV X have CDF $F_X(x)$ and PDF $f_X(x)$.

Let RV $Y = aX + b$, a, b are constants.

What will be $F_Y(y)$ and $f_Y(y)$?

$$F_Y(y) = P(Y \leq y)$$

$$= P(aX + b \leq y)$$

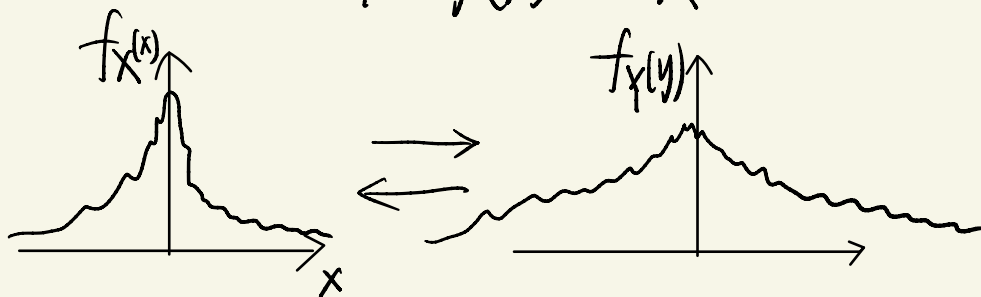
$$= P(aX \leq y - b) \quad \text{Use } F_X(x)$$

$$= \begin{cases} P(X \leq \frac{y-b}{a}), & \text{if } a > 0 \\ P(X \geq \frac{y-b}{a}), & \text{if } a < 0 \end{cases} \quad \downarrow \quad \begin{cases} F_X(\frac{y-b}{a}), & a > 0 \\ 1 - F_X(\frac{y-b}{a}), & a < 0 \end{cases}$$

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = \begin{cases} f_X\left(\frac{y-b}{a}\right) \cdot \frac{1}{a}, & a > 0 \\ -f_X\left(\frac{y-b}{a}\right) \cdot \frac{1}{a}, & a < 0 \end{cases}$$

$$= \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

Visualization: $Y = g(X) = 2X$



Example: X is a Gaussian RV with mean μ and variance σ^2 , and $Y = aX + b$.

Use Special case 2

What is PDF/CDF of Y ?

$$\text{PDF of } Y: f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

PDF of Gaussian

with mean $b + a\mu$, variance $a^2\sigma^2$.

Linear transformation of Gaussian RV is still Gaussian!

$$\begin{aligned} &= \frac{1}{|a|} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}} \\ &= \frac{1}{\sqrt{2\pi} |a| \sigma} e^{-\frac{(y - (b + a\mu))^2}{2\sigma^2 |a|^2}} \end{aligned}$$